Limit: Motivation

A rock breaks loose from the top of a cliff. What is its *average* speed

1. during the first two seconds of fall?
2. during the 1-second interval between second 1 and second 2?

Solution: Galileo's law: The distance fallen is proportional to the square of the time it has been falling.
Indeed if y denotes the distance fallen in feet in t seconds, then

$$y = 16t^2.$$

1. Thus the average speed during the first two seconds is

$$\frac{\Delta y}{\Delta t} = \frac{16(2^2) - 16(0^2)}{2 - 0} = 32 \text{ ft/sec.}$$

2. The average speed during the 1-second interval between second 1 and second 2 is

$$\frac{\Delta y}{\Delta t} = \frac{16(2^2) - 16(1^2)}{2 - 1} = 48 \text{ ft/sec.}$$

Limit: Motivation

Find the instantaneous speed of the rock at $t = 1$ and $t = 2$ seconds.

Solution: The average speed of the rock over a time interval $[t_0, t_0 + h]$ having length h is

$$\frac{\Delta y}{\Delta t} = \frac{16((t_0 + h)^2) - 16(t_0^2)}{h} \text{ ft/sec.}$$

To calculate the speed at t_0, we cannot simply substitute $h = 0$ in the above formula as we cannot divide by zero.
But we can use this formula to compute the the average speed over increasingly short time intervals starting at $t_0 = 1$ and $t_0 = 2$. For $h \neq 0$, the above formula simplifies as follows:

$$\frac{\Delta y}{\Delta t} = \frac{16((t_0 + h)^2) - 16(t_0^2)}{h} = \frac{16(t_0^2 + 2t_0h + h^2) - 16t_0^2}{h} = \frac{32t_0h + 16h^2}{h} = 32t_0 + 16h.$$

Thus the instantaneous speed of the rock at $t_0 = 1$ second is 32 ft/sec and the instantaneous speed of the rock at $t_0 = 2$ second is 64 ft/sec.

Limit: Motivation

Definition

The **average rate of change** of $y = f(x)$ with respect to x over the interval $[x_1, x_2]$ of length $h \neq 0$ is

$$\frac{\Delta y}{\Delta t} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}.$$

Note: Geometrically, the average rate of change of f over $[x_1, x_2]$ is the *slope* of the line through the points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$.
What does happen when $x_2 = x_1$?
We rather see what happens when x_2 approaches x_1 as we cannot substitute x_1 for x_2.
When x_2 approaches x_1, $\frac{\Delta y}{\Delta t} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$ could be approaching a finite value.

Definition (Informal Definition of Limit)

Let $f(x)$ be defined on an open interval containing c except possibly at c. If $f(x)$ is arbitrarily close to the number L (as close as we like) for all x sufficiently close to c, other than c itself, then we say that the **limit of** $f(x)$ **as** x **approaches** c **is the number** L, and write

$$\lim_{x \to c} f(x) = L.$$

In other words, the values of $f(x)$ are close to the number L whenever x is close to c.

Problem. How does the function

$$f(x) = \frac{x^2 - 1}{x - 1}$$

behave near $x = 1$?

Solution: The given formula defines f for all real numbers x except $x = 1$. For $x \neq 1$, the formula simplifies as follows:

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x + 1, \qquad \text{for } x \neq 1.$$

For values of x close to 1, $f(x)$ is close to 2.
In this case, we write

$$\lim_{x \to 1} f(x) = 2.$$

Thus the graph of f is the line $y = x + 1$ with the point $(1, 2)$ removed.

Let

$$g(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 1, & x = 1. \end{cases}$$

What is $\lim_{x \to 1} g(x)$? Is it equal to $g(1)$?

Let

$$h(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 2, & x = 1. \end{cases}$$

What is $\lim_{x \to 1} h(x)$? Is it equal to $h(1)$?

Examples

1. $\lim_{x \to 2} 24 = 24.$
2. $\lim_{x \to 3} 10 = 10.$
3. $\lim_{x \to 4} x^2 = 16.$
4. $\lim_{x \to 3} (3 - 5x) = -12.$
5. $\lim_{x \to -1} \dfrac{3 - 2x}{x - 1} = \dfrac{-5}{2}.$

More Examples

1. If f is the **identity function** $f(x) = x$, then for any value c

$$\lim_{x \to c} f(x) = \lim_{x \to c} x = c.$$

2. If f is the **constant function** $f(x) = k$, then for any value c

$$\lim_{x \to c} f(x) = \lim_{x \to c} k = k.$$

Nonexistence of Limit

Discuss the behavior of the following functions as $x \to 0$:

$$U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

.

Theorem (Limit Laws)

Let L, M, c be real numbers and let

$$\lim_{x\to c} f(x) = L \qquad \text{and} \qquad \lim_{x\to c} g(x) = M. \text{ Then}$$

1. *Sum Rule:* $\lim_{x\to c}(f(x) + g(x)) = L + M.$
2. *Difference Rule:* $\lim_{x\to c}(f(x) - g(x)) = L - M.$
3. *Constant Multiple Rule:* $\lim_{x\to c}(k \cdot f(x)) = k \cdot L.$
4. *Product Rule:* $\lim_{x\to c}(f(x) \cdot g(x)) = L \cdot M.$
5. *Quotient Rule:* $\lim_{x\to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \qquad M \neq 0.$
6. *If r and s are integers with no common factors and $s \neq 0$, then*

$$\lim_{x\to c} f(x)^{r/s} = L^{r/s}$$

provided $L^{r/s}$ is a real number. (If s is even, then $L > 0$.)

Example: Limit Laws

Use the observations $\lim_{x\to c} k = k$ and $\lim_{x\to c} x = c$ and the properties of limits to find the following limits.

1. $\lim_{x\to c}(x^3 + 4x^2 - 3) = \lim_{x\to c} x^3 + \lim_{x\to c} 4x^2 - \lim_{x\to c} 3 = c^3 + 4c^2 - 3.$
2. $\lim_{x\to c} \dfrac{x^4 + x^2 - 1}{x^2 + 5} = \dfrac{\lim_{x\to c}(x^4 + x^2 - 1)}{\lim_{x\to c}(x^2 + 5)} = \dfrac{c^4 + c^2 - 1}{c^2 + 5}.$
3. $\lim_{x\to -2} \sqrt{4x^2 - 3} = \sqrt{\lim_{x\to -2}(4x^2 - 3)} = \sqrt{13}.$

Theorem (Limits of Polynomials)

If $P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_0$, *then*

$$\lim_{x \to c} P(x) = P(c) = a_n c^n + a_{n-1}c^{n-1} + \ldots + a_0.$$

Theorem (Limits of Rational Functions)

If $P(x)$ *and* $Q(x)$ *are polynomials and* $Q(c) \neq 0$, *then*

$$\lim_{x \to c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

Theorem (The Sandwich Theorem for Limits of Functions)

Suppose that $g(x) \leq f(x) \leq h(x)$ *in some open interval containing* c *except possibly at* $x = c$ *itself. Suppose also that*

$$\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L.$$

Then $\lim_{x \to c} f(x) = L$.

Example:
Let

$$1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{4} \quad \text{for all } x \neq 0.$$

Since $\lim_{x \to 0}(1 - \frac{x^2}{4}) = \lim_{x \to 0}(1 + \frac{x^2}{4}) = 1$, by the Sandwich Theorem, $\lim_{x \to 0} u(x) = 1$ no matter how complicated u is.

Examples

1. From the definition of $\sin\theta$, we have that $-|\theta| \leq \sin\theta \leq |\theta|$.
   Also $\lim_{\theta\to 0}(-|\theta|) = \lim_{\theta\to 0}|\theta| = 0$.
   Hence, by the Sandwich Theorem,

$$\lim_{\theta\to 0}\sin\theta = 0.$$

2. From the definition of $\cos\theta$, $0 \leq 1 - \cos\theta \leq |\theta|$.
   Hence $\lim_{\theta\to 0}(1-\cos\theta) = 0$ or

$$\lim_{\theta\to 0}\cos\theta = 1.$$

3. For any function $f(x)$, if $\lim_{x\to c}|f(x)| = 0$, then $\lim_{x\to c} f(x) = 0$ since $-|f(x)| \leq f(x) \leq |f(x)|$ for all x.

Theorem

If $f(x) \leq g(x)$ for all x in some open interval containing c, except possibly at $x = c$, and the limits f and g both exist as x approaches c, then

$$\lim_{x \to c} f(x) \leq \lim_{x \to c} g(x).$$

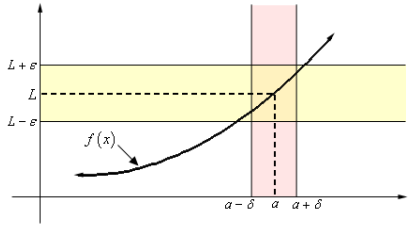
Definition (The Precise Definition of Limit)

Let $f(x)$ be defined on an open interval containing c except possibly at c. We say that the **limit of** $f(x)$ **as** x **approaches** c **is the number** L, and write

$$\lim_{x \to c} f(x) = L,$$

if for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$



Example

Show that

$$\lim_{x \to 1} \left(\frac{3}{2}x - 1 \right) = \frac{1}{2}.$$

Solution: Here $c = 1$, $f(x) = \frac{3}{2}x - 1$ and $L = \frac{1}{2}$.
Let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that

$$0 < |x - 1| < \delta \quad \Rightarrow \quad |f(x) - \frac{1}{2}| < \epsilon.$$

We work backward:

$$\begin{aligned} \left| \left(\frac{3}{2}x - 1 \right) - \frac{1}{2} \right| &= \left| \frac{3}{2}x - \frac{3}{2} \right| < \epsilon \\ &\Leftrightarrow \frac{3}{2}|x - 1| < \epsilon \\ &\Leftrightarrow |x - 1| < \frac{2}{3}\epsilon \end{aligned}$$

Let us take $\delta = \frac{2}{3}\epsilon$. Let $0 < |x-1| < \delta = \frac{2}{3}\epsilon$. Then

$$\left|\left(\frac{3}{2}x - 1\right) - \frac{1}{2}\right| = \left|\frac{3}{2}x - \frac{3}{2}\right| = \frac{3}{2}|x-1| < \frac{3}{2}\left(\frac{2}{3}\epsilon\right) = \epsilon.$$

Thus

$$\lim_{x \to 1}\left(\frac{3}{2}x - 1\right) = \frac{1}{2}.$$

Example

Prove using the definition of limit that (a) $\lim_{x \to c} x = c$ and (b) $\lim_{x \to c} k = k$.

Solution: (a) Let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Rightarrow \quad |x - c| < \epsilon.$$

The above implication will hold if δ equals ϵ or any other positive number less than ϵ. This prove that

$$\lim_{x \to c} x = c.$$

(b) Let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Rightarrow \quad |k - k| < \epsilon.$$

Since $|k - k| = 0$, we can use any positive number as δ and the implication will hold. Thus

$$\lim_{x \to c} k = k.$$

Example

Prove that $\lim_{x \to 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2. \end{cases}$$

Solution: Given $\epsilon > 0$, we must show that there is a $\delta > 0$ such that

$$0 < |x - 2| < \delta \quad \Rightarrow \quad |f(x) - 4| < \epsilon.$$

We first solve the inequality $|f(x) - 4| < \epsilon$ to find an open interval containing $c = 2$ on which the inequality holds for all $x \neq c$:

$$\begin{aligned} |f(x) - 4| < \epsilon \quad &\Leftrightarrow \quad |x^2 - 4| < \epsilon \\ &\Leftrightarrow \quad -\epsilon < x^2 - 4 < \epsilon \\ &\Leftrightarrow \quad 4 - \epsilon < x^2 < 4 + \epsilon \\ &\Leftrightarrow \quad \sqrt{4 - \epsilon} < |x| < \sqrt{4 + \epsilon} \quad \text{assuming } \epsilon \leq 4. \end{aligned}$$

Thus for $x \neq 2$ such that $\sqrt{4-\epsilon} < x < \sqrt{4+\epsilon}$, we have $|f(x) - 4| < \epsilon$.
We have thus found an open interval containing 2 (but excluding 2) for which $|f(x) - 4| < \epsilon$.

We now choose a $\delta > 0$ such that the centered interval $(2-\delta, 2+\delta)$ is contained in the above interval. For this, we take $\delta = \min\{2 - \sqrt{4-\epsilon}, \sqrt{4+\epsilon} - 2\}$. This implies

$$0 < |x - 2| < \delta \quad \Rightarrow \quad |f(x) - 4| < \epsilon.$$

Similarly, if $\epsilon > 4$, then we choose δ in such a way that $(2-\delta, 2+\delta)$ is contained in the interval $(0, \sqrt{4+\epsilon})$. Thus, we choose $\delta = \min\{2, \sqrt{4+\epsilon} - 2\}$. As $|f(x) - 4| < \epsilon$ for all $x \in (0, \sqrt{4+\epsilon})$, it also holds for all $x \in (2-\delta, 2+\delta)$.

This shows that $\lim_{x \Longrightarrow \infty} f(x) = 4$.

Using the Definition of Limit to Prove Theorems

Prove that, if $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$, then

$$\lim_{x \to c} (f(x) + g(x)) = L + M.$$

Solution: Let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Rightarrow \quad |f(x) + g(x) - (L + M)| < \epsilon.$$

Since $\lim_{x \to c} f(x) = L$, corresponding to $\epsilon/2 > 0$, there exists a $\delta_1 > 0$ such that

$$0 < |x - c| < \delta_1 \quad \Rightarrow \quad |f(x) - L| < \epsilon/2.$$

Similarly, since $\lim_{x \to c} g(x) = M$, corresponding to $\epsilon/2 > 0$, there exists a $\delta_2 > 0$ such that

$$0 < |x - c| < \delta_2 \quad \Rightarrow \quad |g(x) - M| < \epsilon/2.$$

Let

$$\delta = \min\{\delta_1, \delta_2\}.$$

Let x be such that $0 < |x - c| < \delta$. Then $0 < |x - c| < \delta_1$ and $0 < |x - c| < \delta_2$ and hence

$$|f(x) + g(x) - (L + M)| = |f(x) - L + g(x) - M| \leq |f(x) - L| + |g(x) - M| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Homework

Prove that, if $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$ and that $f(x) \leq g(x)$ in an open interval containing c except possibly at c, then $L \leq M$.

One-Sided Limits

If $\lim_{x \to c} f(x) = L$, then it means that as x approaches c from either left or from right $f(x)$ approaches L.
In other words, both the left-hand and right-hand limits exist and equal L.
That is,

$$\lim_{x \to c^-} f(x) = L \quad \text{and} \quad \lim_{x \to c^+} f(x) = L.$$

The converse is also true: If the left-hand and right-hand limits of a function exist as x approaches c and are equal, then $\lim_{x \to c} f(x)$ exists and equals the common left-hand and right-hand limits.

Theorem

A function $f(x)$ has a limit as x approaches c if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x \to c} f(x) = L \iff \lim_{x \to c-} f(x) = L \quad \textit{and} \quad \lim_{x \to c^+} f(x) = L.$$

One-Sided Limits: Examples

1. The function $f(x) = \dfrac{x}{|x|}$ has limit 1 as x approaches 0 from the right and has limit -1 as x approaches 0 from the left. Thus the left-hand and right-hand limits exists as x approaches 0 but they are not equal. So, $\lim\limits_{x \to 0} f(x)$ does not exist.
2. The function $f(x) = \sqrt{4 - x^2}$ is defined at all points on the closed interval $[-2, 2]$. The limit of $f(x)$ exist at all points in the open interval $(-2, 2)$. We also have

$$\lim_{x \to -2^+} \sqrt{4 - x^2} = 0 \quad \text{and} \quad \lim_{x \to 2^-} \sqrt{4 - x^2} = 0.$$

This function does not have a left-hand limit at $x = -2$ and a right-hand limit at $x = 2$. Thus it does not have ordinary two-sided limits at either -2 or 2.

The Precise Definition

Definition (Right-Hand Limit)

We say that $f(x)$ has **right-hand limit** L **at** c, and write

$$\lim_{x \to c^+} f(x) = L,$$

if for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$c < x < c + \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

Definition (Left-Hand Limit)

We say that $f(x)$ has **left-hand limit** L **at** c, and write

$$\lim_{x \to c^-} f(x) = L,$$

if for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$c - \delta < x < c \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

Example

Prove using the definition of limit that

$$\lim_{x \to 0^+} \sqrt{x} = 0.$$

Solution: Let $\epsilon > 0$ be given. Here $c = 0$ and $L = 0$. So, we want to find a $\delta > 0$ such that

$$0 < x < \delta \quad \Rightarrow \quad |\sqrt{x} - 0| < \epsilon \quad \text{or} \quad 0 < x < \delta \quad \Rightarrow \quad \sqrt{x} < \epsilon.$$

Now $\sqrt{x} < \epsilon$ if $0 < x < \epsilon^2$. So, we take $\delta = \epsilon^2$. And have that

$$0 < x < \delta = \epsilon^2 \quad \Rightarrow \quad \sqrt{x} < \epsilon.$$

This shows that $\lim_{x \to 0^+} \sqrt{x} = 0$.

Homework

1. Show that $f(x) = \sin(\frac{1}{x})$ has no limit as x approaches zero from either side.
2. Prove that $\lim_{x \to 0} \frac{\sin \theta}{\theta} = 1$.

Continuity

Definition (Continuous at a Point)

A function $y = f(x)$ is **continuous at an interior point** c of its domain if

$$\lim_{x \to c} f(x) = f(c).$$

A function $y = f(x)$ is **continuous at a left end point point** a of its domain if

$$\lim_{x \to a^+} f(x) = f(a).$$

A function $y = f(x)$ is **continuous at a right end point point** b of its domain if

$$\lim_{x \to b^-} f(x) = f(b).$$

Examples

1. The identity function $f(x) = x$ is continuous at every point in the interval $(-\infty, \infty)$.
2. The constant function $f(x) = k$ is continuous at every point in the interval $(-\infty, \infty)$.
3. The function $f(x) = \dfrac{1}{x}$ is continuous at every point in the interval $(0, \infty)$.
4. The function $f(x) = \sqrt{4 - x^2}$ is continuous at every point in the interval $[-2, 2]$. In particular, it is **right-continuous** at $x = -2$ and is **left-continuous** at $x = 2$.
5. The unit step function $U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$ is continuous at all points except 0. It is right-continuous at $x = 0$ but is not left-continuous there. So, it is not continuous at $x = 0$. It has a jump discontinuity at $x = 0$.

Homework: Discuss the continuity of the greatest integer function $f(x) = \lfloor x \rfloor$; that is, $f(x)$ is the greatest integer less than or equal to x for every x. (It is right-continuous at every integer n but is not left-continuous.)

Types of Discontinuity

Definition

- A function $f(x)$ is said to have a **removable discontinuity** at $x = c$ if $\lim_{x \to c} f(x)$ exists but f is not defined at c or $f(c)$ is defined at c but is not equal to the limit at c.
- A function $f(x)$ is said to have a **jump discontinuity** at $x = c$ if the left-hand and right-hand limits exist at $x = c$ but they are not equal.
- A function $f(x)$ is said to have an **infinite discontinuity** at $x = c$ if the left-hand and/or the right-hand limits at c are $\pm\infty$.
- A function $f(x)$ is said to have an **oscillating discontinuity** at $x = c$ if the function oscillates too much as x approaches c and so does not have a limit at $x = c$.

Examples

- Removable discontinuity: The function $f(x) = \begin{cases} x+3, & x \neq 0 \\ 1, & x = 0 \end{cases}$ has a removal discontinuity at $x = 2$.
- Jump discontinuity: The unit step function $U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$ has a jump discontinuity at $x = 0$.
- Infinite discontinuity: The function $f(x) = \dfrac{1}{x^2}$ has an infinite discontinuity at $x = 0$.
- Oscillating discontinuity: The function $f(x) = \sin(\frac{1}{x})$ has an oscillating discontinuity at $x = 0$.

Definition

A function $f(x)$ is said to be **continuous on an interval** if it is continuous at every point of the interval.

Examples:

- The identity function $f(x) = x$ is continuous on the interval $(-\infty, \infty)$.
- The constant function $f(x) = k$ is continuous on the interval $(-\infty, \infty)$.
- The function $f(x) = \sqrt{4 - x^2}$ is continuous on the interval $[-2, 2]$.
- The function $f(x) = \sqrt{x}$ is continuous on the interval $[0, \infty)$.
- The function $f(x) = 1/x$ is continuous on the interval $(0, \infty)$ but is not continuous on the interval $(-\infty, \infty)$.

Theorem

If f and g are continuous at $x = c$, then the following combinations are also continuous at $x = c$:

1. *Sums:* $f + g$.
2. *Differences:* $f - g$.
3. *Constant Multiples:* $k \cdot f$.
4. *Products:* $f \cdot g$.
5. *Quotients:* $\frac{f}{g}$.
6. *Powers:* $f^{r/s}$, *provided it is defined on an open interval containing c, where r and s are integers.*

Proof:

The theorem follows from the theorem on limits of sum of functions, difference of functions, etc. For instance, we prove Part (1) as follows:

$$\begin{aligned}
\lim_{x \to c} (f+g)(x) &= \lim_{x \to c} (f(x) + g(x)) \\
&= \lim_{x \to c} f(x) + \lim_{x \to c} g(x) \\
&= f(c) + g(c) \\
&= (f+g)(c)
\end{aligned}$$

Examples

1. Every polynomial $P(x)$ is continuous because $\lim_{x \to c} P(x) = P(c)$.
2. Every rational function $\dfrac{P(x)}{Q(x)}$ is continuous wherever it is defined (i.e., $Q(c) \neq 0$).
3. The absolute value function $f(x) = |x|$ is continuous at every value of x. If $x > 0$, $f(x) = x$, a polynomial. If $x < 0$, $f(x) = -x$, another polynomial. At the origin, $\lim_{x \to 0} |x| = 0 = |0|$.

Note

A function $f(x)$ is continuous at $x = c$ if and only if

$$\lim_{x \to c} f(x) = f(c) \quad \text{or} \quad \lim_{h \to 0} f(c + h) = f(c).$$

Homework: Prove that $\sin x$ and $\cos x$ are continuous at every x. (Use the facts that $\lim_{x \to 0} \sin x = 0$ and $\lim_{x \to 0} \cos x = 1$.)

The Continuity of Composite Functions

Theorem

If f is continuous at c and g continuous at $f(c)$, then the composite function $g \circ f$ is continuous at c.

Homework: Show that the following functions are continuous wherever they are defined.

1. $f(x) = \sqrt{x^2 - 2x - 5}$.
2. $f(x) = \left| \dfrac{x-2}{x^2-2} \right|$.

Continuous Extension to a Point

The function $f(x) = \dfrac{\sin x}{x}$ is defined and continuous at every point except $x = 0$. But it has a finite limit, namely 1, as $x \to 0$. So, we define a new function

$$F(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

$F(x)$ is defined and continuous at every x. In particular, it is continuous at $x = 0$ because

$$\lim_{x \to 0} F(x) = \lim_{x \to 0} \frac{\sin x}{x} = 1 = F(0).$$

$F(x)$ is called the **continuous extension of** $f(x)$ **to** $x = c$.

Homework: Show that

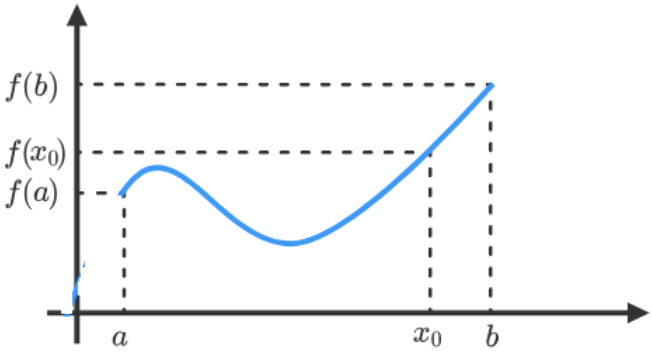
$$f(x) = \frac{x^2 + x - 2}{x^2 - 1}$$

has a continuous extension to $x = 1$ and find the extension.

Theorem (Intermediate Value Theorem for Continuous Functions)

A function $y = f(x)$ that is continuous on a closed interval $[a, b]$ takes on every value between $f(a)$ and $f(b)$.

In other words, if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.



Note: The image is from the internet.

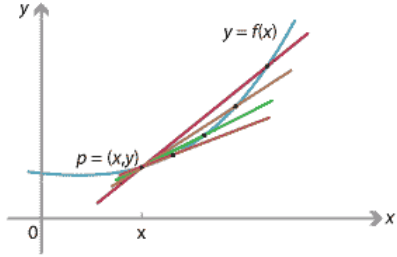
Differentiation

1. It is one of the main motivations for the concept of limit.
2. It tell us a how a curve bends at a point on the curve, by providing the slope of the curve at that point.
3. It gives us the rate of change of a function at a point.
4. It helps us to define many physical concepts such as velocity, acceleration, etc with ease.

The Slope of a Curve, Informally

Consider a curve $y = f(x)$. Let P be a point on the curve. What is the slope of the curve at P? What is the tangent line to the curve at P?

1. Start with a neighbouring point Q on the curve.
2. Investigate the limit of the secant slope as Q approaches P along the curve.
3. If the limit exists, take it as the **slope** of the curve at P.
4. And take the line through this P with the slope computed as the **tangent line** to the curve at P.



Note: The image is from the internet.

Example

Find the slope of the parabola $y = x^2$ at the point $P(2, 4)$. Write an equation for the tangent to the parabola at this point.

Solution: We begin with a secant line through $P(2, 4)$ and $Q(2 + h, (2 + h)^2)$ nearby on the curve. (h may be positive or negative.)
The slope of this secant line is

$$\frac{\Delta y}{\Delta x} = \frac{(2+h)^2 - 2^2}{h} = \frac{h^2 + 4h + 4 - 4}{h} = \frac{h^2 + 4h}{h} = h + 4.$$

Thus, as Q approaches P along the curve, h approaches 0 and the secant slope approaches 4:

$$\lim_{h \to 0} h + 4 = 4.$$

So, we take 4 as the parabola's slope at P. The tangent to the parabola at P is the line through P with slope 4:

$$y = 4 + 4(x - 2) \quad \text{or} \quad y = 4x - 4.$$

Definition (Slope, Tangent Line)

The **slope of the curve** $y = f(x)$ at the point $P(c, f(c))$ is the number

$$m = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \quad \text{(provided the limit exists).}$$

The tangent line to the curve at P is the line through P with this slope.

Example

Show that the line $y = mx + b$ is its own tangent at any point $(c, mc + b)$.

Solution: Let $f(x) = mx + b$. So, $f(c) = mc + b$ and $f(c + h) = m(c + h) + b = mc + mh + b$.

Then, the slope at $(c, f(c))$ is

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0} \frac{(mc + mh + b) - (mc + b)}{h} = \lim_{h \to 0} \frac{mh}{h} = m.$$

Hence the tangent line is

$$y = (mc + b) + m(x - c)$$

or

$$y = mc + b + mx - mc$$

or

$$y = mx + b.$$

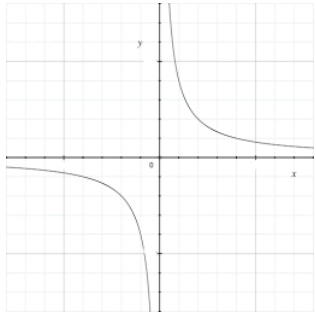
Example

Find the slope of the curve $y = 1/x$ at $x = a \neq 0$. What happens to the tangent to the curve at the point $(a, 1/a)$ as a changes?

Solution: Here $f(x) = 1/x$. Thus, the slope at $(a, 1/a)$ is

$$\begin{aligned} m &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \to 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h \to 0} \frac{1}{h} \, \frac{a - (a+h)}{a(a+h)} \\ &= \lim_{h \to 0} \frac{-1}{a(a+h)} \\ &= -\frac{1}{a^2}. \end{aligned}$$

Example: Slope and Tangent to the Curve $y = 1/x, x \neq 0$



- The slope at $x = a \neq 0$, $-\dfrac{1}{a^2}$, is always negative.
- As a approaches 0 from either direction, the slope approaches $-\infty$ and the tangent becomes increasingly steep.
- As a moves away from the origin in either direction, the slope approaches 0 and tangent becomes increasingly horizontal.

Definition (Derivative at a Point)

The expression

$$\frac{f(c+h)-f(c)}{h}$$

is called the **difference quotient of** f **at** c **with increment** h. If the difference quotient has a limit as h approaches 0, that limit is called the **derivative of** f **at** c.

In other words, the **derivative of a function** $f(x)$ at a point c is defined as

$$f'(c) = \lim_{h \to 0} \frac{f(c+h)-f(c)}{h},$$

provide the limit exists.

Note:

1. If we interpret the difference quotient as a secant slope, the derivative gives the slope of the curve and tangent at the point $x = c$.
2. If we interpret the difference quotient as an average rate of change, the derivative gives the function's rate of change with respect to x at the point $x = c$. (Recall the falling rock example.)

We have defined the derivative of a function $f(x)$ at a point c. We now define the derivative function:

Definition (Derivative Function)

The **derivative** of the function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provide the limit exists.

Alternatively,

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}.$$

Definition (Differentiability, Differentiation)

1. For a function $f(x)$, if $f'(x)$ exists at a particular point x, we say that f is **differentiable (or has a derivative)** at x.
2. If f' exists at every point in the domain of f, we say that the function f **is differentiable**.
3. The process of calculating a derivative is called **differentiation**.
4. The derivative is alternatively denoted as

$$f'(x) = \frac{d}{dx} f(x).$$

Examples

We have seen that the derivative of $y = mx + b$ at any point x is m. Thus

$$\frac{d}{dx}(mx + b) = m.$$

For instance,

$$\frac{d}{dx}\left(\frac{3}{2}x - 4\right) = \frac{3}{2}.$$

We have seen that the derivative of $y = 1/x, x \neq 0$, at any $x \neq 0$ is $-1/x^2$. Thus

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}.$$

Example

Differentiate $f(x) = \dfrac{x}{x-1}$.

Solution: Here $f(x) = \dfrac{x}{x-1}$ and $f(x+h) = \dfrac{x+h}{x+h-1}$. So,

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{\frac{x+h}{x+h-1} - \frac{x}{x-1}}{h} \\
&= \lim_{h\to 0} \frac{1}{h} \frac{(x+h)(x-1) - x(x+h-1)}{(x+h-1)(x-1)} \\
&= \lim_{h\to 0} \frac{1}{h} \frac{-h}{(x+h-1)(x-1)} \\
&= \lim_{h\to 0} \frac{-1}{(x+h-1)(x-1)} \\
&= \frac{-1}{(x-1)^2}.
\end{aligned}$$

Example

(a) Find the derivative of $y = \sqrt{x}$ for $x > 0$. (b) Also find the tangent line to this curve at $x = 4$.

Solution: (a) We use the alternative formula to calculate derivative:

$$\begin{aligned} f'(x) &= \lim_{z \to x} \frac{f(z) - f(x)}{z - x} \\ &= \lim_{z \to x} \frac{\sqrt{z} - \sqrt{x}}{z - x} \\ &= \lim_{z \to x} \frac{\sqrt{z} - \sqrt{x}}{(\sqrt{z} - \sqrt{x})(\sqrt{z} + \sqrt{x})} \\ &= \lim_{z \to x} \frac{1}{\sqrt{z} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}}. \end{aligned}$$

Example

(b) The slope of the curve at $x = 4$ is

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

Thus the tangent at $x = 4$ is the line through the point $(4, 2)$ with slope $1/4$:

$$y = 2 + \frac{1}{4}(x - 4) \quad \text{or} \quad y = \frac{1}{4}x + 1.$$

Notation

There are many notations for the derivative of a function $y = f(x)$ with respect to x:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = D(f)(x) = D_x(f(x)).$$

Definition (One-Sided Derivatives)

The **right-hand derivative** of a function $f(x)$ at a point $x = c$ is the limit

$$\lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h} \quad \text{(provided the limit exists).}$$

The **left-hand derivative** of a function $f(x)$ at a point $x = c$ is the limit

$$\lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \quad \text{(provided the limit exists).}$$

Fact. A function $f(x)$ has a derivative at c if and only if it has left-hand and right-hand derivatives at c and these one-sided derivatives are equal.

Definition (Differentiable on an Interval)

A function $y = f(x)$ is **differentiable** on an open interval (finite or infinite) if it has a derivative at each point of the interval.

It is **differentiable** on a closed interval $[a, b]$ if it is differentiable on the interior (a, b) and has right-hand derivative at a and left-hand derivative at b.

Examples

Show that the function $f(x) = |x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$ but has no derivative at $x = 0$.

Solution: For $x > 0$,

$$\frac{d}{dx}(|x|) = \frac{d}{dx}(x) = \frac{d}{dx}(1 \cdot x + 0) = 1.$$

For $x < 0$,

$$\frac{d}{dx}(|x|) = \frac{d}{dx}(-x) = \frac{d}{dx}(-1 \cdot x + 0) = -1.$$

We show that $|x|$ is not differentiable at $x = 0$ by showing that the one-sided derivatives differ there:

$$\begin{aligned}
\text{Right-hand derivative of } |x| \text{ at zero} &= \lim_{h \to 0^+} \frac{|0+h| - |0|}{h} \\
&= \lim_{h \to 0^+} \frac{h}{h} \\
&= \lim_{h \to 0^+} 1 = 1.
\end{aligned}$$

$$\begin{aligned}
\text{Left-hand derivative of } |x| \text{ at zero} &= \lim_{h \to 0^-} \frac{|0+h| - |0|}{h} \\
&= \lim_{h \to 0^+} \frac{-h}{h} \\
&= \lim_{h \to 0^+} -1 = -1.
\end{aligned}$$

Homework

Show that $y = \sqrt{x}$ is not differentiable at $x = 0$. (Show that the right-hand derivative does not exist at $x = 0$.)

Differentiable Functions are Continuous

Theorem

If f is differentiable at $x = c$, then f is continuous at $x = c$.

Proof Suppose f is differentiable at $x = c$: i.e., $f'(c)$ exists.

To show that f is continuous at $x = c$, we must show that $\lim_{x \to c} f(x) = f(c)$ or, equivalently, that $\lim_{h \to 0} f(c + h) = f(c)$.

For $h \neq 0$,

$$f(c + h) = f(c) + (f(c + h) - f(c)) = f(c) + \frac{f(c + h) - f(c)}{h} \cdot h.$$

Now take limit as $h \to 0$:

$$\lim_{h \to 0} f(c + h) = \lim_{h \to 0} f(c) + \lim_{h \to 0} \frac{f(c + h) - f(c)}{h} \cdot \lim_{h \to 0} h = f(c) + f'(c) \cdot 0 = f(c).$$

The Intermediate Value Property of Derivatives

Theorem

If a and b are any two points in an interval on which f is differentiable, then f' takes on every value between $f'(a)$ and $f'(b)$.

If a and b are any two points in an interval I with $a < b$, then for every y between $f'(a)$ and $f'(b)$, there exists an $x \in [a, b]$ such that $f'(x) = y$.

Application: The unit step function cannot be the derivative of any function.

Differentiation: Some Simple Results

1. If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

2. If n is a positive integer, then

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Proof of (2):

$$\begin{aligned}
\frac{d}{dx}x^n &= \lim_{z\to x}\frac{z^n - x^n}{z - x} \\
&= \lim_{z\to x}\frac{(z-x)(z^{n-1} + z^{n-2}x + z^{n-3}x^2 + \cdots + x^{n-1})}{z - x} \\
&= \lim_{z\to x}(z^{n-1} + z^{n-2}x + z^{n-3}x^2 + \cdots + x^{n-1}) \\
&= nx^{n-1}.
\end{aligned}$$

Differentiation Rules

1. **Constant Multiple Rule:** If u is a differentiable function of x and c is a constant, then
$$\frac{d}{dx}(cu) = c\frac{du}{dx}.$$
2. **Sum Rule:** If u and v are differentiable functions of x, then their sum $u + v$ is differentiable at every point where both u and v are differentiable. At such cases,
$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}.$$
3. **Sum Rule for More Than Two Functions:** If $u_1, u_2, \ldots, u_n$ are differentiable functions of x, then their sum $u_1 + u_2 + \cdots + u_n$ is differentiable at every point where all of $u_1, u_2, \ldots, u_n$ are differentiable. At such cases,
$$\frac{d}{dx}(u_1 + \cdots + u_n) = \frac{du_1}{dx} + \cdots + \frac{du_n}{dx}.$$

Example

Does the curve $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so where?

Solution: The horizontal tangents, if any, occur where the slope $\frac{dy}{dx}$ is zero.

Now

$$\frac{dy}{dx} = \frac{d}{dx}(x^4 - 2x^2 + 2) = 4x^3 - 4x.$$

$$\begin{aligned} &\frac{dy}{dx} = 0 \\ \iff &4x^3 - 4x = 0 \\ \iff &x(x^2 - 1) = 0. \\ \iff &x = 0, 1, -1. \end{aligned}$$

Thus the curve has horizontal tangents at $(0, 2), (1, 1)$ and $(-1, 2)$ on the curve.

More Differentiation Rules

1. If u and v are differentiable at x, then so is their product uv and
$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$
2. If u and v are differentiable at x and if $v(x) \neq 0$, then the quotient u/v is differentiable at x and
$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

Power Rule for Negative Integers

If n is a negative integer and $x \neq 0$, then

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Proof. Let $n = -m$, where m is a positive integer. Then

$$\begin{aligned}
\frac{d}{dx}(x^n) &= \frac{d}{dx}\left(\frac{1}{x^m}\right) \\
&= \frac{x^m \frac{d}{dx}(1) - 1\frac{d}{dx}(x^m)}{(x^m)^2} \\
&= \frac{0 - mx^{m-1}}{x^{2m}} \\
&= -mx^{-m-1} \\
&= nx^{n-1}.
\end{aligned}$$

Homework

1. Find an equation for the tangent to the curve

$$y = x + \frac{2}{x}$$

at the point $(1, 3)$.

2. Find the derivative of

$$y = \frac{(x-1)(x^2-2x)}{x^4}.$$

Second and Higher Order Derivatives

If $y = f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function.

If f' is also differentiable, we can differentiate f' to get a new function of x denoted by f''. So,

$$f'' = (f')'.$$

The function f'' is called the **second derivative** of f.

Notations:

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dy'}{dx} = y'' = D^2(f)(x) = D_x^2 f(x).$$

More generally, the ***n*th derivative** of $y = f(x)$ is the derivative of the $(n-1)$th derivative of f and is denoted by

$$f^{(n)} = y^{(n)} = \frac{d}{dx}y^{(n-1)} = \frac{d^ny}{dx^n} = D^n y.$$

Homework

Find the first four derivatives of $y = x^3 - 3x^2 + 2$.

The Chain Rule

If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

Alternatively, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Homework

An object moves along the x-axis so that its position at any time $t \geq 0$ is given by $x(t) = \cos(t^2 + 1)$. Find the velocity of the object as a function of t.

Parametric Formula for dy/dx

If all three derivatives exist and $dx/dt \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

Homework: Consider the curve described by a particle whose position $P(x, y)$ at time t is given by

$$x = a\cos t, \quad y = b\sin t, \quad 0 \leq t \leq 2\pi.$$

Find the line tangent to the curve at the point $(a/\sqrt{2}, b/\sqrt{2})$, where $t = \pi/4$.

Implicit Differentiation: An example

If p/q is a rational number, then $x^{p/q}$ is differentiable at every point where $x^{(p/q)-1}$ is defined and

$$\frac{d}{dx}x^{p/q} = \frac{p}{q}x^{(p/q)-1}.$$

Proof: Let p and q be integers with $q > 0$ and suppose that $y = x^{p/q}$. Then

$$y^q = x^p.$$

Differentiating both sides with respect to x, we get

$$qy^{q-1}\frac{dy}{dx} = px^{p-1}.$$

We divide both sides of the equation by qy^{q-1} and obtain

$$
\begin{aligned}
\frac{dy}{dx} &= \frac{px^{p-1}}{qy^{q-1}} \\
&= \frac{p}{q} \cdot \frac{x^{p-1}}{\left(x^{p/q}\right)^{q-1}} \\
&= \frac{p}{q} \cdot \frac{x^{p-1}}{x^{p-p/q}} \\
&= \frac{p}{q} \cdot x^{(p-1)-(p-p/q)} \\
&= \frac{p}{q} \cdot x^{(p/q)-1}.
\end{aligned}
$$

Definition (Linearization, Standard Linear Approximation)

If f is differentiable at $x = a$, then the approximating function

$$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of f at a.

The approximation

$$f(x) \approx L(x)$$

of f by L is called the **standard linear approximation** of f at a.

Example

Find the linearization of $f(x) = \sqrt{1+x}$ at $x = 0$.

Solution: Since

$$f'(x) = \frac{1}{2}(1+x)^{-1/2},$$

we have $f(0) = 1$ and $f'(0) = 1/2$. Thus the linearization at $x = 0$ is

$$L(x) = f(a) + f'(a)(x-a) = 1 + \frac{1}{2}(x-0) = 1 + \frac{x}{2}.$$

Definition (Differentials)

Let $y = f(x)$ be a differentiable function. The **differential** dx is an independent variable. The **differential** dy is

$$dy = f'(x)dx.$$

Example. (a) Find dy if $y = x^5 + 37x$. (b) Find the value of dy when $x = 1$ and $dx = 0.2$.

Solution: (a) $dy = (5x^4 + 37)dx$.

(b) Substituting $x = 1$ and $dx = 0.2$ in the expression for dy, we obtain

$$dy = (5 \cdot 1^4 + 37)(0.2) = 8.4.$$

Absolute Extrema

Definition (Absolute Maximum, Absolute Minimum)

Let f be a function with domain D. Then f has an **absolute maximum** value on D at a point c if

$$f(x) \leq f(c) \qquad \text{for all } x \text{ in } D$$

and an **absolute minimum** value on D at a point c if

$$f(x) \geq f(c) \qquad \text{for all } x \text{ in } D.$$

Examples

Consider the function $y = x^2$. Absolute extrema depend on the domain D:

1. If $D = (-\infty, \infty)$, then the function has no absolute maximum. But it has absoute minimum of 0 at $x = 0$.
2. If $D = [0, 2]$, then the function has absolute maximum of 4 at $x = 2$ and absolute minimum of 0 at $x = 0$.
3. If $D = (0, 2]$, then the function has absolute maximum of 4 at $x = 2$ but it has no absolute minimum.
4. If If $D = (0, 2)$, then the function has no absolute extrema.

Theorem (The Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$.

That is, there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = m$ and $f(x_2) = M$ and $m \leq f(x) \leq M$ for every other x in $[a, b]$.

Local Extreme Values

Definition (Local Maximum, Local Minimum)

A function f has a **local maximum** value at an interior point c of its domain if

$$f(x) \leq f(c) \qquad \text{for all } x \text{ in some open interval containing } c.$$

A function f has a **local minimum** value at an interior point c of its domain if

$$f(x) \geq f(c) \qquad \text{for all } x \text{ in some open interval containing } c.$$

Theorem (The First Derivative Theorem for Local Extreme Values)

If f has a local maximum or minimum value at an interior point c of its domain and if f' is defined at c, then

$$f'(c) = 0.$$

Proof:

- Suppose f has a local maximum at an interior point $x = c$. So $f(x) - f(c) \leq 0$ for all values of x close enough to c.
- Suppose $f'(c)$ is defined. Since c is an interior point, it then follows this derivative is defined by the two-sided limit
$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$
- This means that the left-hand and right-hand derivatives exist at c and both equal $f'(c)$.
- So, we have
$$f'(c) = \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} \leq 0$$
and
$$f'(c) = \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0.$$
- These inequalities imply that $f'(c) = 0$.

Note: The converse of this theorem is not true: $y = x^3$ has derivative 0 at $x = 0$ but it is not a point of local extremum.

Definition (Critical Point)

An interior point of the domain of a function f where f' is zero or undefined is a critical point.

How to Find the Absolute Extrema of a Continuous Function f on a finite closed interval:

1. Evaluate f at all critical points and endpoints.
2. Take the largest and smallest of these values.

Example

Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2, 1]$.

Solution: The function is differentiable over the entire domain. So the only critical point is where

$$f'(x) = 2x = 0; \quad \text{i.e } x = 0.$$

Thus we must check the function f at the interior point $x = 0$ and at the end points $x = -2$ and $x = 1$:

Critical point value: $f(0) = 0$.

Endpoint values: $f(-2) = 4$ and $f(1) = 1$.

The function has an absolute maximum value of 4 at $x = -2$ and an absolute minimum value of 0 at $x = 0$.

Homework

1. Find the absolute extrema values of $g(t) = 8t - t^4$ on $[-2, 1]$.
2. Find the absolute extrema values of $f(x) = x^{2/3}$ on the interval $[-2, 3]$.

Theorem (Rolle's Theorem)

Suppose $y = f(x)$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b). If

$$f(a) = f(b),$$

then there is at least one number c in (a, b) at which

$$f'(c) = 0.$$

Proof:

- Since f is continuous on $[a, b]$, by the extreme value theorem, it assumes absolute maximum and minimum values on $[a, b]$.
- These can occur (i) at interior points where f' is zero; (ii) at interior points where f' does not exist; (iii) at the endpoints a and b.
- By hypothesis, f' exists at every interior point. So, the second possibility is ruled out.
- If the minimum or maximum occur at an interior point c, then $f'(c) = 0$ (by the first derivative test) and the Rolle's theorem follows.
- If both the absolute maximum and absolute minimum occur at the end points, then since $f(a) = f(b)$ it follows that $f(x) = f(a) = f(b)$ for all x in $[a, b]$. So, it follows that $f'(x) = 0$ for all x in (a, b) and thus we can take any c from the interior (a, b).

Example

The function

$$f(x) = \frac{x^3}{3} - 3x$$

is

- continuous at every point of $[-3, 3]$
- differentiable at every point of $(-3, 3)$
- and $f(-3) = f(3) = 0$.

Thus by Rolle's theorem there is at least one point c in $(-3, 3)$ where f' is zero.

In fact, there are two points, namely $x = -\sqrt{3}$ and $x = \sqrt{3}$ for which f' is zero.

Example

Show that the equation

$$x^3 + 3x + 1 = 0$$

has exactly one real solution.

Solution:

- Let
$$f(x) = x^3 + 3x + 1.$$
- If the equation $x^3 + 3x + 1 = 0$ has (at least) two real roots at the points $x = a$ and $x = b$, then by Rolle's theorem, there is an interior point c between a and b such that $f'(c) = 0$.
- But, the derivative
$$f'(x) = 3x^2 + 3$$
is never zero, because it is always positive.
- Thus, the function $f(x)$ has at most one real root.
- In fact, it has exactly one real root, that follows from the intermediate value theorem as $f(-1) = -3 < 0$ and $f(0) = 1 > 0$.

Theorem (The Mean Value Theorem)

Suppose $y = f(x)$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b). Then there is at least one point c in (a, b) at which

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Proof:

- Consider the line through the points $A(a, f(a))$ and $B(b, f(b))$. It is the graph of the function
$$g(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$
- The vertical difference between the graphs f and g at x is
$$h(x) = f(x) - g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$
- The function $h(x)$ satisfies the hypothesis of Rolle's theorem on $[a, b]$, thus, there is a point c in (a, b) where $h'(c) = 0$.
- But
$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$
- So, $h'(c) = 0$ implies that $f'(c) - \frac{f(b) - f(a)}{b - a} = 0$. That is
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Corollaries

Corollary

If $f'(x) = 0$ at each point x of an open interval (a, b), then $f(x) = C$ for all x in (a, b), where C is a constant.

Corollary

If $f'(x) = g'(x)$ at each point x of an open interval (a, b), then there exists a constant C such that $f(x) = g(x) + C$ for all x in (a, b). That is, $f - g$ is a constant.